

Diffusion of the Tracer

The tracer spreads according to the diffusion equation below.
The coefficient D was measured for each temperature.

$$dC/dt = D * d^2C/dx^2 + k * C$$

$$D = D_0 * \exp(-E_a / (R * T))$$

Here C is the concentration and t is time in seconds.
The activation energy E_a was 42.5 kJ per mole in every run.